Trabajo Final Ingles 3

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Abstract:

Predictive Model of Industrial Production by Sector in Tierra del Fuego

This project presents a machine learning model developed to predict monthly industrial production across various sectors in Tierra del Fuego. The purpose of the study is to support data-driven decisions by analyzing production behavior in industries that report output in different units, such as kilograms or physical units. Historical data on production volume, employment, and establishments were collected, cleaned, and merged into time series. Then, sectors were grouped by output type, and two general models were trained separately. Ridge Regression was used for unit-based sectors, achieving an R^2 of 0.918 with strong consistency in cross-validation (0.872). ElasticNet was applied to kilogram-based sectors, obtaining an R^2 of 0.688 and robust performance. Although some individual models had limited validation success, the general models showed better balance between accuracy and stability. As a result, the units model was used to predict Electronics and Confectionist production, and the kilograms model was applied to Textiles and Fisheries. These findings suggest that grouping sectors by measurement unit enhances predictive reliability. This approach can assist policymakers and industry leaders in anticipating sectoral trends, optimizing resource allocation, and guiding strategic decisions based on historical production data.